COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	**CPE 010 - DATA STRUCTURES AND ALGORITHMS**	**SECTION:**	CPE21S4
DATE PERFORMED:	9 / 18 / 26	**DATE SUBMITTED:**	9 / 18 / 26
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Activity NO. 9.1 **Tree ADT**

1. PROCEDURE OUTPUT

What procedures did you perform?

```cpp
#include <iostream>
#include "BST.h"

int main (){
    //create the root node:
    BinarySearchTree<int> tree(20);

    tree.insert(10);
    tree.insert(5);
    tree.insert(70);
    tree.insert(80);
    tree.insert(40);
    tree.insert(50);

    std::cout<<"preOrder:";
    tree.preOrder();
    std::cout<<std::endl;

    std::cout<<"inOrder:";
    tree.inOrder();
    std::cout<<std::endl;

    std::cout<<"postOrder:";
    tree.postOrder();
    std::cout<<std::endl;

    return 0;
}
```

```cpp
#ifndef BST_H
#define BST_H

template <typename T>
class BinarySearchTree{
    private:
        T key;
        //pointers:
        BinarySearchTree<T>*left;
        BinarySearchTree<T>*right;



    public:
        //constructor
        BinarySearchTree<T>(T value);
        void insert (T value);

        //traversal algo:
        void preOrder();
        void inOrder();
        void postOrder();
};
template <typename T>
BinarySearchTree<T>::BinarySearchTree(T value){
    key = value;
    left = nullptr;
    right = nullptr;
}

template <typename T>
void BinarySearchTree<T>::insert(T value){
    if (value < key){
        if(left == nullptr){
            left = new BinarySearchTree<T>(value);
        }else{
```

```
38              left->insert(value);
39          }
40      }
41      else if (value > key){
42          if(right == nullptr){
43              right = new BinarySearchTree<T>(value);
44          }else{
45              right ->insert(value);
46          }
47      }
48  }
49  template <typename T>
50  void BinarySearchTree<T>::preOrder(){
51      std::cout<<key<<" ";
52      if (left != nullptr){
53          left->preOrder();
54      }
55      if (right != nullptr){
56          right->preOrder();
57      }
58
59  }
60
61  template <typename T>
62  void BinarySearchTree<T>::inOrder(){
63
64      //go to left
65      if (left != nullptr){
66          left->inOrder();
67      }
68      std::cout<<key<<" ";
69      //go to right
```

```
70      if (right != nullptr){
71          right->inOrder();
72      }
73
74  }
75
76  template <typename T>
77  void BinarySearchTree<T>::postOrder(){
78
79      //go to left
80      if (left != nullptr){
81          left->postOrder();
82      }
83
84      //go to right
85      if (right != nullptr){
86          right->postOrder();
87      }
88
89      std::cout<<key<<" ";
90  }
91  #endif
```

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used:

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use:

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use:

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the

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